

MT 509: Hydrogen Energy

Course Code	MT 509
Course Name	Hydrogen Energy
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	B.Tech./PG/Ph.D.
Mutual Exclusion	Some similarly with CY552: Hydrogen Generation and Storage
Approval	57th BoA (Oct 2024)

Course Contents

- **Introduction to Hydrogen Energy and Production:** Properties of hydrogen, global status of supply and demand, and key hydrogen production methods (steam reforming, partial oxidation, autothermal reforming). Comparison of conventional and advanced methods, and production from hydrocarbons and biomass. (6 Hours)
- **Hydrogen Production from Renewable Sources:** Thermochemical cycles for hydrogen production, electrolysis of water (fundamentals, electrolytic cell design, stack configuration), photoelectrochemical hydrogen production, and a comparative technical and economic analysis of various production methods. (6 Hours)
- **Hydrogen Storage:** Introduction to hydrogen storage methods, fundamentals of hydrogen compression, liquefaction, and underground storage. Hydrogen storage in tanks and adsorption-based materials, metal hydrides, and design of hydrogen storage systems. (6 Hours)
- **Transportation and Distribution of Hydrogen:** Long-distance hydrogen transport via pipelines, ships, and as liquid organic hydrogen carriers (LOHC). Road transportation of hydrogen and refueling stations. Economic analysis of transport methods. (4 Hours)
- **Utilization of Hydrogen:** Hydrogen use in internal combustion engines, fuel cells, and other energy conversion devices. Applications of hydrogen in different sectors, with a focus on decarbonization. (6 Hours)
- **Hydrogen Safety and Regulations:** Properties of hydrogen associated with hazards, safety concerns in storage and transport, classification of hazards, safety regulations, codes, and standards. Case studies of hydrogen incidents and mitigation strategies. (6 Hours)

Laboratory Modules

1. NA

Text Books

1. Gupta, Ram B., **Hydrogen Fuel: Production, Transport and Storage**, CRC Press, 2008.

References

1. Tzimas, E., Filiou, C., Peteves, S.D., & Veyret, J.B., **Hydrogen storage: state-of-the-art and future perspective**, European Communities, 2003.
2. Agata Godula-Jopek, **Hydrogen Production by Electrolysis**, Wiley-VCH, 2015
3. Global Hydrogen Review 2021, IEA (2021), Paris, <https://www.iea.org/reports/global-hydrogen-review-2021>
4. Michael Hirscher, **Handbook of Hydrogen Storage**, Wiley-VCH, 2010.
5. Yuda Yurum (Editor), **Hydrogen energy system: production and utilization of hydrogen and future aspects**, Springer Science & Business Media; NATO ASI Series, Series E: Applied Sciences – Vol. 295, 2012.